姓名: SOLUTION　　葉均承　　應數一線性代數

學號: _____　　**Quiz 7**　　考試日期: 2026/10/28

不可使用手機、計算器，禁止作弊!

1. Given $A \sim H$, please answer the following questions.

$$A = \begin{bmatrix} 6 & 2 & 0 & -3 & -2 \\ -2 & 4 & 14 & 6 & -7 \\ 4 & -2 & -10 & 3 & 7 \\ 8 & 0 & -8 & 9 & 7 \\ 2 & 3 & 7 & -6 & -7 \end{bmatrix}, H = \begin{bmatrix} 2 & 0 & -2 & 0 & 1 \\ 0 & 1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) the **rank** of matrix A, is **3** .

(b) Is A invertible? **NO!** .

(c) a basis for the **row space** of A is **[2, 0, -2, 0, 1], [0, 1, 3, 0, -2], [0, 0, 0, 3, 1]** .

(d) a basis for the **column space** of A is $\begin{bmatrix} 6 \\ -2 \\ 4 \\ 8 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ -2 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 6 \\ 3 \\ 9 \\ -6 \end{bmatrix}$.

(e) a basis for the **nullspace** of A is $\left\{ \begin{bmatrix} 1 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 2 \\ 0 \\ -1/3 \\ 1 \end{bmatrix} \right\}$ **or** $\left\{ \begin{bmatrix} 1 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 12 \\ 0 \\ -2 \\ 6 \end{bmatrix} \right\}$.

Solution：

(a) There's 3 pivots in matrix H.

(b) Pick the rows in **H** which contains a pivot.

(c) Pick the columns in **A** which the corresponding columns in H contains a pivot, i.e., the 1st, 2nd and 4th columns of A.

(d) Let $x_3 = r, x_5 = s$. By **H**, $2x_1 - 2x_3 + x_5 = 0, x_2 + 3x_3 - 2x_5 = 0, 3x_4 + x_5 = 0$. Thus $x_1 = r - \frac{1}{2}s, x_2 = -3r + 2s, x_4 = -\frac{1}{3}s$.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = r \begin{bmatrix} 1 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1/2 \\ 2 \\ 0 \\ -1/3 \\ 1 \end{bmatrix}$$

2. Prove or disprove (反證) the following statement.

   (a) The column space of AC is contained in the column space of A.

   Solution :

   It is true! 2-2, problem 14.

   (b) $rank(AC) \leq rank(A)$.

   Solution :

   It is true! 2-2, problem 18.

   (c) The column space of AC is contained in the column space of C.

   Solution :

   It is false! 2-2, problem 15.

   (d) $rank(AC) \leq rank(C)$.

   Solution :

   It is true! 2-2, problem 20.